

		1.40 J to 1.76 J
6(a)(i)	Total resistance $= 100 + \left(\frac{200 \times 6000}{200 + 6000} \right) + 300$ $= 593.548 \, \Omega$ $\approx 590 \, \Omega$	[1] for correct substitution [1] for correct answer
6(a)(ii)	Current from battery $= \frac{12}{593.548}$ $= 0.020217 \, \text{A}$ $\approx 0.020 \, \text{A}$	[1] for correct substitution and correct answer

6(a)(iii)	Effective resistance of $200\ \Omega$ resistor and voltmeter $= \frac{200 \times 6000}{200 + 6000} = 193.548\ \Omega$ Voltmeter reading $= \left(\frac{193.548}{100 + 193.548 + 300} \right) 12$ $= 3.913\ \text{V}$ $\approx 3.9\ \text{V}$ OR Voltmeter reading $= (0.020217)(193.548)$ $= 3.913\ \text{V}$ $\approx 3.9\ \text{V}$	[1] for effective resistance [1] for correct substitution
6(b)	With light intensity at a maximum, the resistance of the LDR is $120\ \Omega$. Effective resistance of $100\ \Omega$ resistor and LDR $= \frac{100 \times 120}{100 + 120} = 54.545\ \Omega$ p.d. across $200\ \Omega$ resistor $= \left(\frac{193.548}{54.545 + 193.548 + 300} \right) 12$ $= 4.238\ \text{V}$ Power supplied to $200\ \Omega$ resistor $= \frac{4.238^2}{200}$ $= 0.0898\ \text{W}$ $\approx 0.090\ \text{W}$	[1] for p.d. across $200\ \Omega$ resistor [1] for correct substitution [1] for correct answer
7(a)	Magnetic flux through a plane surface is defined as the product	[1] or zero